RAVI YADAV

Software Developer — API Development | Integration | Unit Testing | Debugging

Nepal
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PROFESSIONAL SUMMARY

Software developer and Bachelor of Computer Applications (BCA) graduate (Lovely Professional University, 2020, CGPA 7.3/10) applying for the Developer role at SDOC. Experience implementing application features and APIs from requirements, developing and optimising database queries and application logic with validation and error handling, writing unit and integration tests, participating in code review, and investigating defects and production issues through to reviewed fixes. Builder of four live open-source products and an active contributor to public projects — including a fix cherry-picked into the Graphify 0.9.95 release on PyPI. Comfortable maintaining technical documentation and supporting deployment verification and handover. Fluent in English, Nepali, Hindi and Bhojpuri.

CORE SKILLS — MATCHED TO THIS ROLE

Programming	Python, Dart (Flutter), Kotlin, HTML/CSS, JavaScript, Visual Basic
API & Integration	REST API development and integration, payment/QR integration (EMVCo, NEPALPAY/FonePay), idempotent transaction design, JSON-based services
Databases	MySQL, SQLite, Oracle Database, MS Access — schema design, query development and optimisation, validation and error handling
Testing & Debugging	Unit and integration testing, regression tests, defect and production-issue investigation, addressing QA and security findings
Tools & Practices	Git, GitHub, code review, technical documentation, release preparation, Microsoft Office (Word, Excel, PowerPoint, Access)
Additional	Frontend and backend web development, PC and online-server troubleshooting, SEO/digital tools (Google AdSense, Trends)

PROJECT & OPEN-SOURCE EXPERIENCE

Independent Developer & Open-Source Contributor — 2019 to Present

Self-directed product development and contributions to public open-source projects; all work is publicly verifiable on GitHub (github.com/08raviyadav, account since 2019).

- **NepPay** — Nepal payment-app prototype (Flutter, Android-first, Node.js backend): implemented an EMVCo QR parser for NEPALPAY/FonePay, a payment adapter with idempotency so retries never double-charge, QR pay, send-to-VPA with name inquiry, request/split payments and history, on a double-entry ledger; documented architecture and security practices in the public repository.
- **SkillVet** — Zero-dependency single-file Python security scanner for AI-agent skills/MCP configurations (27 rules, 7 categories) with scoring, JSON output, CI exit codes and a GitHub Action; shipped with full documentation (usage, rules, FAQ, security policy) and a passing test suite.
- **CtrlZ** — Filesystem-level undo tool for AI-agent file changes (Python CLI: snapshot/undo/redo/restore) using a content-addressed blob store; verified with a disaster-recovery test restoring deleted and overwritten files byte-for-byte.
- **Aura Scanner** — Offline Android app (Kotlin, Jetpack Compose) with photo/name input, animated scan flow, scored results and sharing; prepared with Play Store listing and privacy policy.
- **Open-source contributions** — Graphify: root-cause fix for MySQL/MariaDB column defaults breaking SQL parsing and silently dropping CREATE TABLE statements — cherry-picked by the maintainer into Graphify 0.9.95 on PyPI, authorship preserved. Further reviewed fix PRs under consideration in FastMCP (OAuth redirect-URI validation — security), LangGraph (configuration validation, 27 new tests), ColossalAI (unsafe checkpoint loading — security) and nanobot (session/file handling, full test suite passing), each delivered with regression tests and verification evidence.

Earlier Academic Projects

- **IoT Gesture-Controlled Laptop** (Jan–May 2020, team of 3) — Arduino UNO and sensors to control a laptop by hand gestures.

- **Student Management System** (May–Jul 2019) — Python/SQLite; separate teacher/student interfaces for attendance, CGPA and fee records.
- **Student Exam Seating System** (Jan–May 2019) — Visual Basic + MS Access; automated seat assignment by roll number, name or performance.

EDUCATION

Bachelor of Computer Applications (BCA) — Lovely Professional University, Phagwara, Punjab, India — 2020 · CGPA 7.3/10

LANGUAGES

English · Nepali · Hindi · Bhojpuri

INTERESTS

Technology news, reading, travel, swimming.